

Technical Documentation: Causal Forest Analysis for the ASPREE Trial

1. Problem Statement

Traditional randomized controlled trial (RCT) analysis focuses on the Average Treatment Effect (ATE), which assumes treatment homogeneity. However, clinical interventions often exhibit Heterogeneous Treatment Effects (HTE), where certain subgroups may experience a significant benefit while others may experience null effects or even harm. The objective was to utilize a Causal Forest to estimate individualized treatment effects—Conditional Average Treatment Effect (CATE)—within the ASPREE U.S. cohort to move toward precision geriatric medicine.

2. Methodology: The Causal Forest Architecture

The Causal Forest is a non-parametric machine learning method designed for causal inference rather than simple prediction.

- **Objective Function:** Unlike standard random forests that minimize prediction error, the Causal Forest partitions the data by maximizing the variance in the treatment effect estimate between child nodes.
- **Feature Set:** We utilized 28 baseline clinical measures, including physical metrics (grip strength, gait speed, BMI) and cognitive metrics (3MS, CES-D).
- **Algorithm Tuning:**
 - **Honest Splitting:** To ensure valid asymptotic inference, the model utilized honest splitting, separating data used to determine the tree structure from the data used to estimate the treatment effect.
 - **Tree Constraints:** To prevent overfitting, the forest was limited to 100 trees, with each tree pruned to a maximum of 10 terminal nodes.
 - **Class Imbalance Management:** Given the 10% event rate for the primary composite outcome, an augmented training set and stratified sampling were utilized to prevent biased partitioning.

3. The Causal Inference Pipeline

The implementation of the Causal Forest followed a rigorous technical pipeline:

- **Data Partitioning:** We employed stratified sampling to divide the U.S. cohort into training/validation ($n=1,141$) and testing ($n=1,150$) sets, ensuring endpoint balance.

- **Estimation:** The model generated CATE scores for each individual, quantifying the expected difference in the probability of the primary composite outcome if assigned to aspirin versus placebo.
- **Heterogeneity Assessment:** We performed Wald Chi-Squared tests on interaction terms within subsequent proportional hazard models to validate if the discovered HTE was statistically significant on a relative scale.

4. Key Findings

- **Predictive Capability:** The Causal Forest demonstrated robust ability to rank patients by their potential for treatment response, achieving an AUC of 0.74. This benchmark validates the model's discriminatory power in identifying subpopulations with high versus low potential for clinical benefit.
- **Absolute Scale Discovery:** While HTE was not detected on the relative scale (e.g., Hazard Ratios), the model revealed critical heterogeneity on the absolute scale.
- **Clinical Impact:** Participants in the highest-risk quintile identified by the Causal Forest experienced a 13.7% Absolute Risk Reduction (ARR) (95% CI 3.1% to 24.4%) in the primary composite outcome at the 5-year follow-up when treated with aspirin, compared to the placebo group.

5. Conclusion

The Causal Forest successfully bypassed the limitation of "average" treatment effects by identifying a high-risk subgroup where aspirin therapy provided clear clinical benefit. This methodology highlights the power of low-bias, non-parametric approaches in clinical research, demonstrating that machine learning can uncover hidden, clinically significant treatment effects in medium-sized cohorts.